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Hiring Manager

The Johns Hopkins University

Dear Hiring Manager,

My name is Ajay Venkatesh. I'm finishing my M.S. in Computer Engineering at NYU, with production software experience at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus work at NYU's Novel AI Technologies. I'm writing about the Research Assistant role on the BodyMaps program at Johns Hopkins.

The combination of medical AI research, three-dimensional anatomical modeling, and clinical AI deployment is where my graduate work and applied AI experience intersect. My coursework spans **Machine Learning, Deep Learning, Big Data, and Computer Architecture**, and I've worked on **RoBERTa fine-tuning with custom LoRA, ResNet** image classifiers, and computer-vision pipelines using **YOLOv8** and **EfficientNet** for medical image classification.

On healthcare AI specifically: at NYU's Novel AI Technologies I architected an **NSF-funded** AI-powered health platform for iOS and Android, ingesting real-time wearable telemetry and deploying cloud-hosted AI models for health-metric analysis. At PwC I shipped an **NLP-based OCR neural model** on Azure for structured data extraction from clinical documents. The combination of model development and production deployment for clinical settings is the shape of work the BodyMaps program describes.

On research collaboration: I've partnered with researchers at NYU on the NSF-funded platform across the full research-to-production cycle. I'm comfortable contributing to research with senior mentors, writing reproducible code, and translating prototype models into deployable systems.

Stack-wise: **Python, PyTorch, TensorFlow, Hugging Face, scikit-learn**, plus **AWS SageMaker** and **Bedrock** for cloud-based ML deployment, and **Docker** for reproducible environments. Comfortable with medical imaging conventions and the data-handling discipline clinical AI demands.

What pulls me toward BodyMaps is the mission. Building three-dimensional anatomical atlases to advance cancer detection sits at the intersection of careful engineering and consequential medicine, where research outcomes compound into clinical impact for real patients.

I'd welcome the chance to discuss further. Thanks for considering my application.

Sincerely,

Ajay Venkatesh